

Quiz-Strings-Midterms Exercise

String Pattern Analysis and Text Processing in C++

Introduction:

This exercise focuses on analyzing textual data using C++ strings and vectors. Students will implement functions that examine properties of words, identify patterns inside strings, and compute statistics about collections of words. The program processes lists of words, evaluates specific string patterns, and generates summaries of the analyzed data.

Learning Outcomes:

- Implement functions that analyze patterns within strings.
- Use vectors to store and process collections of words.
- Apply string operations such as substring extraction and comparison.
- Compute statistics from groups of textual data.

Problem Description:

```
struct wordRecord
{
    string word;
    int length;
    int vowelCount;
};
```

```
/* Determines the number of vowels contained in a word.
```

```
    The function examines each character of the string and
    counts the total number of vowels present. */
```

```
int countVowels(string w);
```

```
/* Determines whether a word is a palindrome.
```

```
    A palindrome is a word that reads the same forward
    and backward. The function returns true if the word
    satisfies this property. */
```

```
bool isPalindrome(string w);
```

```
/* Creates and returns a wordRecord structure containing
   the word, its length, and its vowel count. */

wordRecord analyzeWord(string w);

/* Processes a vector of strings and returns a vector
   of wordRecord structures representing the analyzed
   information for each word. */

vector<wordRecord> analyzeList(vector<string> words);

/* Computes the average length of the words stored
   inside a vector of wordRecord structures. */

double averageLength(vector<wordRecord> records);
```

Expected Output (partial):

Word Analysis Report

Word: level
Length: 5
Vowels: 2
Palindrome: Yes

Word: computer
Length: 8
Vowels: 3
Palindrome: No

Word: radar
Length: 5
Vowels: 2
Palindrome: Yes

Average Length: 6.0